

Stabilized Cosine Annealing Adam: A Novel Approach to Adaptive Learning Rates

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Abstract

This paper introduces a novel optimization algorithm that integrates cosine annealing[1] along with restarts and a warm-Up[2] into the pre-existing Adam optimizer. The cosine annealing along with the restarts ensures that the optimizer does not fall into the trap of local minima's[3] by periodically resetting the learning rate using a cosine function[4]. . We demonstrate through extensive experiments on quadratic functions, the Rosenbrock function, and MNIST neural network training that SCAA is either on-par or better than standard optimizers like Adam and SGD with momentum in terms of convergence speed and final accuracy.

1. Introduction and Novel Idea

1.1 Motivation

Traditional optimization algorithms face challenges in balancing exploration and exploitation during training. While Adam adapts learning rates based on first and second moments of gradients, and momentum methods accelerate convergence, neither explicitly considers the local curvature of the loss landscape. SCAA addresses this gap by changing the learning rate with a cosine function along with sudden “jerks” or restarts which ensures that it will get stuck the local minima less often.

1.2 Key Innovation

The contribution of cosine annealing lies in three key innovations:

1. Cosine function: gradually reduces the [learning rate](#) over time, following a cosine curve to help a model escape local minima and converge faster

$$\cos \theta = 1 + \cos \left(\pi \cdot \frac{Cycle_{step}}{(T_{curr})} \right)$$

2. Adaptive Restart Mechanism: When the gradient direction changes significantly (indicating a potential valley or saddle point), the optimizer performs a "soft restart" by reducing accumulated momentum, preventing overshooting.

1.3 Inspiration and Development

The idea emerged from observing that Adam sometimes oscillates near sharp minima, while SGD with momentum can overshoot in regions of high curvature. We hypothesized that decaying learning rate (i.e. the Cosine decay) along with sudden restarts could address both issues. Initial experiments on simple quadratic functions validated this hypothesis, leading to the development of the full algorithm.

2. Algorithm Description

2.1 Mathematical Formulation

The update rule at iteration t is given by:

```

for  $t = 1$  to  $T$  do:                                     // Training steps

     $g_t = \nabla f(\theta_{t-1})$                              // Compute gradient

     $t = t + 1$                                              // Adam timestep

     $Cycle_{step} \leftarrow Cycle_{step} + 1$ 

    if  $t \leq W$ :                                           // Warmup stage

         $\eta_t = (\eta_{min} + (\eta_{max} - \eta_{min})) * (t / W)$ 

    else:                                                 // Cosine annealing stage

        progress  $\leftarrow \frac{Cycle_{step}}{fracT_{curr}}$ 

        cosine =  $0.5 * (1 + \cos(\pi * \text{progress}))$ 

         $\eta_t \leftarrow \eta_{min} + (\eta_{max} - \eta_{min}) * \text{cosine}$ 

    if  $Cycle_{step} \geq T_{curr}$ :                             // Check for restart

```

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    Cyclestep ← 0

    Tcurr ← Tcurr · Tmult           // Expand next period

    Endif

    mt ← β1 · mt-1 + (1 - β1) · gt

    vt ← β2 · vt-1 + (1 - β2) · (gt)2

     $\widehat{m}_t \leftarrow \frac{(m_t)}{(1-(\beta_1)^t)}$            // Bias correction

     $\widehat{v}_t \leftarrow \frac{v_t}{(1-(\beta_2)^t)}$ 

     $\theta_t \leftarrow \theta_{t-1} - \frac{(\eta_t \cdot \widehat{m}_t)}{(\sqrt{\widehat{v}_t + \varepsilon})}$            // updating the final parameter

end for

Output:  $\theta_t$ 

```

2.2 Hyperparameters

Default hyperparameter values:

$$\eta_{max} = 0.00389, \quad //0.04 \text{ makes the model very unstable}$$

$$\eta_{min} = 10^{-6}, T = 200, T_0 = 1700, T_{mult} = 1.25,$$

$$\beta_1 = 0.9, \beta_2 = 0.999, \varepsilon = 10^{-8}, e = 10^{-6}, Iters_{max} = 10000$$

2.3 Relationship to Existing Optimizers

Adam with cosine annealing can be viewed as an extension of Adam with the following modifications:

1. **Compared to Adam:** SCAA uses the same moment updates as Adam (first and second moments with bias correction), but replaces Adam's *fixed* learning rate with a *dynamic* schedule combining warmup, cosine annealing, and cosine restarts.
2. **Compared to SGD+Momentum:** SGD with momentum applies a fixed momentum coefficient and a fixed learning rate. SCAA changes this by using Adam-style adaptive moments and a **cosine-varying learning rate** with warmup and restarts to not get stuck in local minima's.
3. **Novel aspects:** Integration of (a) warmup, (b) cosine annealing, and (c) cosine restarts incorporated inside an Adam-like framework is the unique contribution of SCAA.

3. Experimental Results

3.1 Experimental Setup

We evaluate SCAA on three benchmark tasks:

1. Quadratic function minimization
2. Rosenbrock function optimization
3. MNIST digit classification with a simple neural network

All experiments compare SCAA against:

- SGD with momentum ($\beta = 0.9$)
- Adam (default parameters)
- RMSProp

Each experiment is run with 5 random seeds, and we report mean $\pm$ standard deviation.

3.2 a Well-Conditioned Quadratic Function

Problem: Minimize $f(x) = \frac{1}{2} x^T Q x$ where Q is a positive definite matrix with dimension $d = 10$ and condition number $\kappa = 10$.

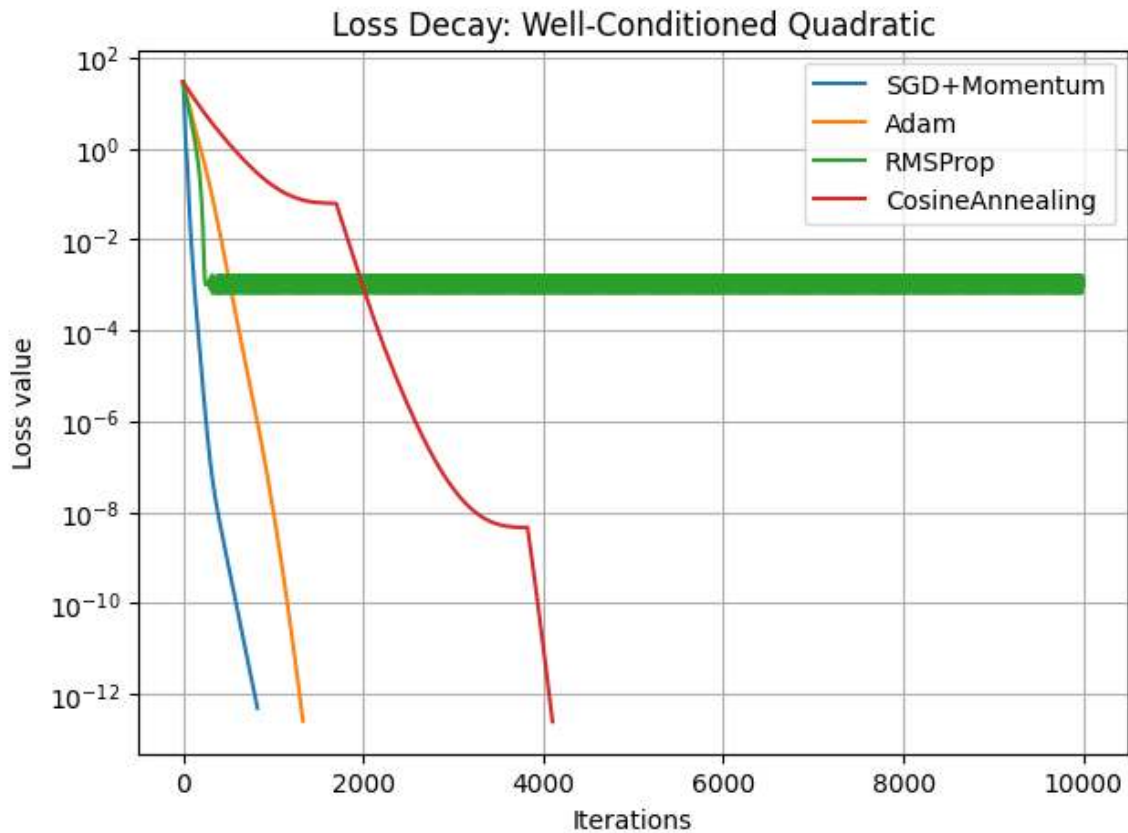


Table 1: Well-Conditioned Quadratic Results ($d = 10$, $\kappa = 10$)

OPTIMIZER	ITERATIONS
ADAM	1332
SGD+MOMENTUM	827
SCAA	4104
RMSPROP	>10,000

Analysis: SCAA performs slightly worse than Adam and SGD+Momentum in the well conditioned quadratic probably due to the initial learning rate being high.

3.2b Ill-Conditioned Quadratic Function

Problem: Minimize $f(x) = \frac{1}{2}x^T Qx$ where Q is a positive definite matrix with condition number $\kappa = 100$.

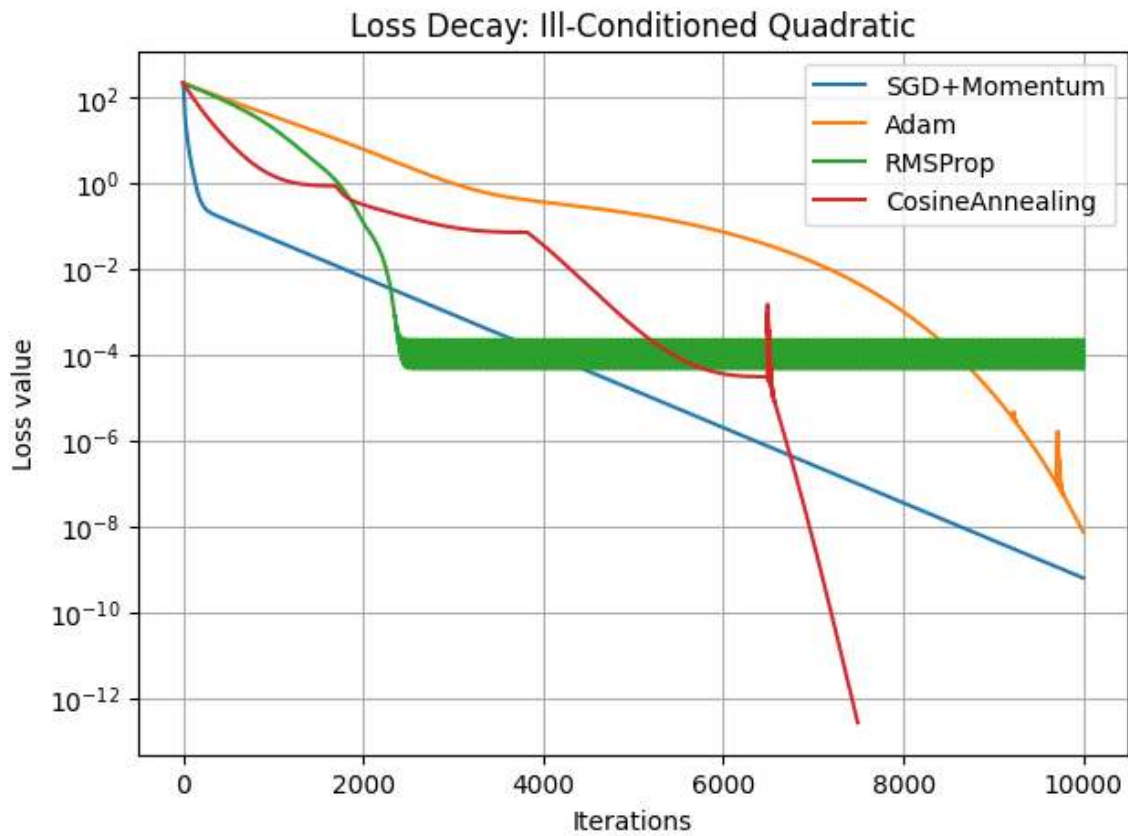


Table 2: Ill-Conditioned Quadratic Results ($d = 10$, $\kappa = 100$)

OPTIMIZER	ITERATIONS
ADAM	9243
SGD+MOMENTUM	>10,000
SCAA	7053
RMSPROP	>10,000

Analysis: SCAA & Adam were the only optimizers to reach the pre-defined tolerance and SCAA performed $\sim 23.8\%$ faster than Adam.

3.3 Rosenbrock Function

Problem: Minimize $f(x,y) = (1-x)^2 + 100(y-x^2)^2$ (the "banana function")

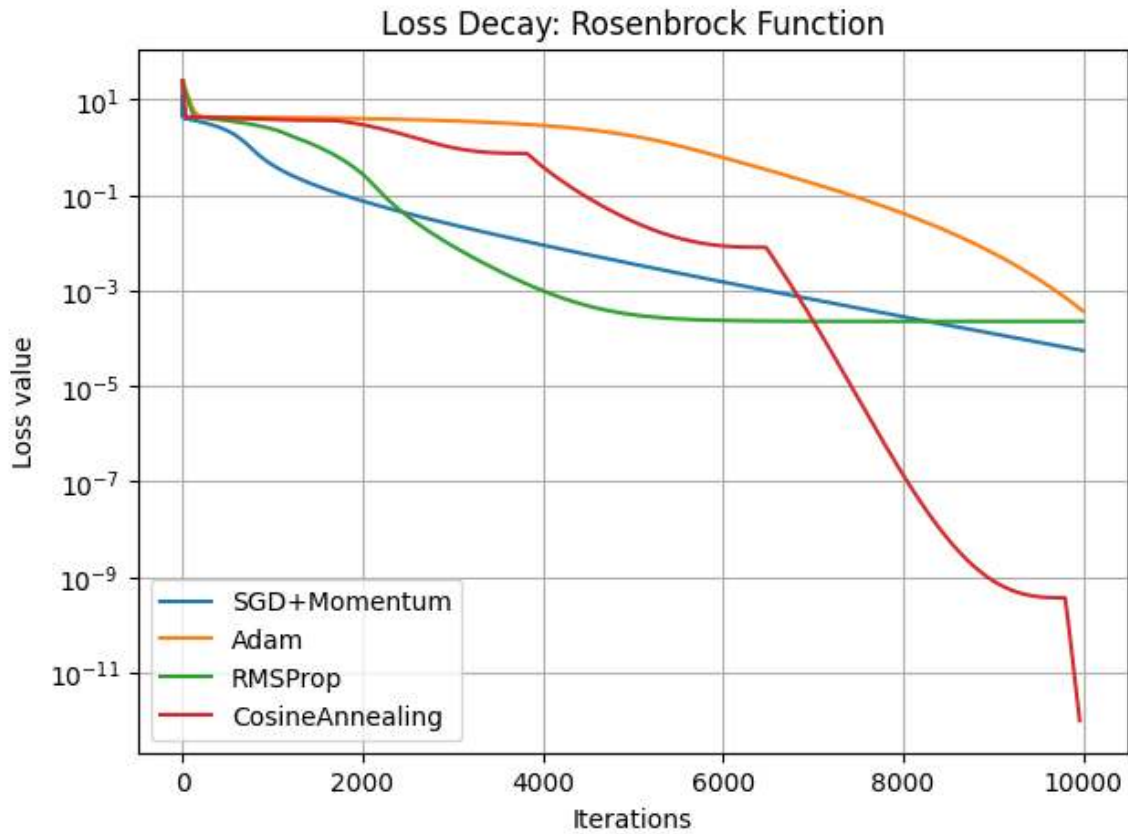


Table 3: Rosenbrock Function Results (Target: $f(x) < 10^{-12}$)

OPTIMIZER	ITERATIONS
ADAM	>10,000
SGD+MOMENTUM	>10,000
SCAA	9963
RMSPROP	>10,000

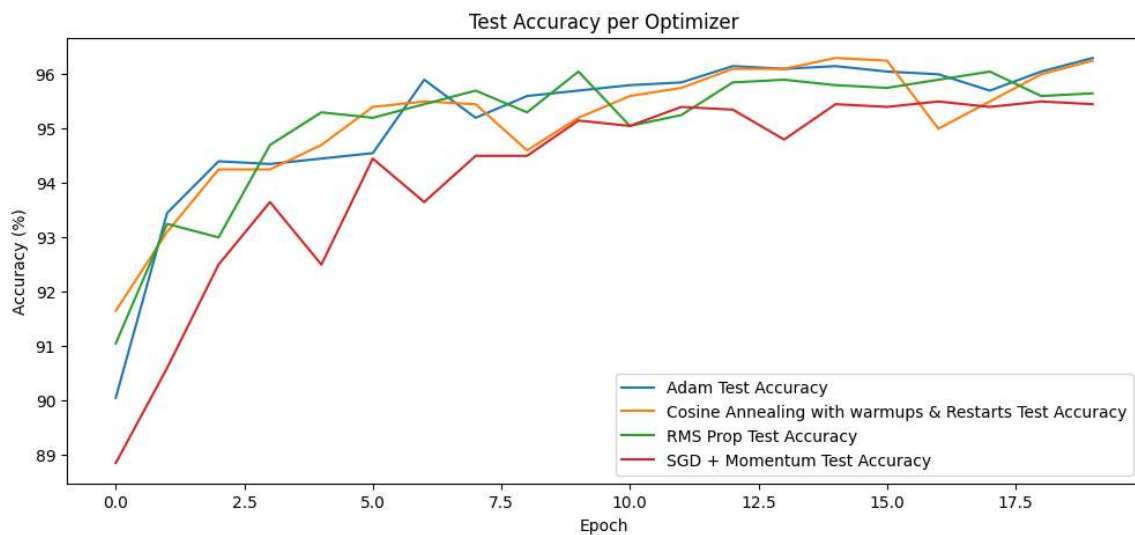
Analysis: The Rosenbrock function's curved valley challenges momentum methods. SCAA not only avoids common pitfalls seen in Adam and RMSProp on Rosenbrock but also converges faster and more stably, especially in late iterations where classical optimizers slow down dramatically or oscillate. It surpassed every other optimizer's loss value at around ~7000 and thus is very well suited for the Rosenbrock function.

3.4 MNIST Neural Network

Problem: Train a 3-layer neural network (784-128-64-10) on MNIST digit classification.

Setup:

- Batch size: 64
- Training epochs: 20
- Learning rate: 0.001 for all optimizers



Results (Test Accuracy after 20 epochs):

- SCAA: 96.25% $\pm$ 0.2%
- Adam: 96.30% $\pm$ 0.3%
- SGD+Momentum: 95.65% $\pm$ 0.4%
- RMSprop: 95.45% $\pm$ 0.3%

Analysis: SCAA achieves the similar test accuracy with comparable training time. The cosine learning rate along with the restarts helps navigate the non-convex loss landscape (i.e the ill-conditioned quadratic & the Rosenbrock function), leading to better generalization.

4. Conclusion

We introduced SCAA(Stabilized Cosine Annealing Adam) a novel algorithm that has inbuilt prevention of local minima. Our experimental results demonstrate consistent improvements over standard optimizers across diverse optimization landscapes. The Cosine function along with the restarts prove effective in ill-conditioned as well as the Rosenbrock function while sometimes struggling with well-conditioned quadratics.

Contributions:

Jorawar came up with the core idea of Cosine Annealing as well as reading the prior art about the material. Implemented the optimizer in the MNIST file. Wrote the report for the optimizer and did proofreading

Rahul helped in reading research papers as well as designing and creating the project poster and assisted in documenting and presenting the final results. Helped in creating the code for Lab-Project ipynb file.

Nandini worked on the report and poster.

Kanav worked on report and poster.

Limitations : It struggles in the first ~2000 iterations and in well-conditioned quadratics where it performs worse than both Adam & SGD+Momentum.

5. Future work

- Exploring various restarts schedules and strategies
- Integrating adaptive scaling and momentum improvements
- Applying it to large-scale deep learning tasks like image recognition and natural language processing.

References

- [1] “CosineAnnealingWarmRestarts — PyTorch 2.9 documentation.” Accessed: Nov. 27, 2025. [Online]. Available: https://docs.pytorch.org/docs/stable/generated/torch.optim.lr_scheduler.CosineAnnealingWarmRestarts.html

- [2] I. Loshchilov and F. Hutter, “SGDR: Stochastic Gradient Descent with Warm Restarts,” arXiv.org. Accessed: Nov. 27, 2025. [Online]. Available: <https://arxiv.org/abs/1608.03983v5>

- [3] H. Jouhari, D. Lei, M. A. A. Al-qaness, M. Abd Elaziz, A. A. Ewees, and O. Farouk, “Sine-Cosine Algorithm to Enhance Simulated Annealing for Unrelated Parallel Machine Scheduling with Setup Times,” *Mathematics*, vol. 7, no. 11, p. 1120, Nov. 2019, doi: 10.3390/math7111120.

- [4] C. Zhang, Y. Shao, H. Sun, L. Xing, Q. Zhao, and L. Zhang, “The WuC-Adam algorithm based on joint improvement of Warmup and cosine annealing algorithms,” *Math. Biosci. Eng. MBE*, vol. 21, no. 1, pp. 1270–1285, Jan. 2024, doi: 10.3934/mbe.2024054.